Sales Prediction using Python

CodeAlpha Data Science Internship | Task 4

1. Project Overview

This project develops regression models to estimate sales using advertising expenditure across TV, Radio, and Newspaper channels. The supplied Advertising dataset is used for data inspection, exploratory analysis, model training, and evaluation.

2. Dataset Description

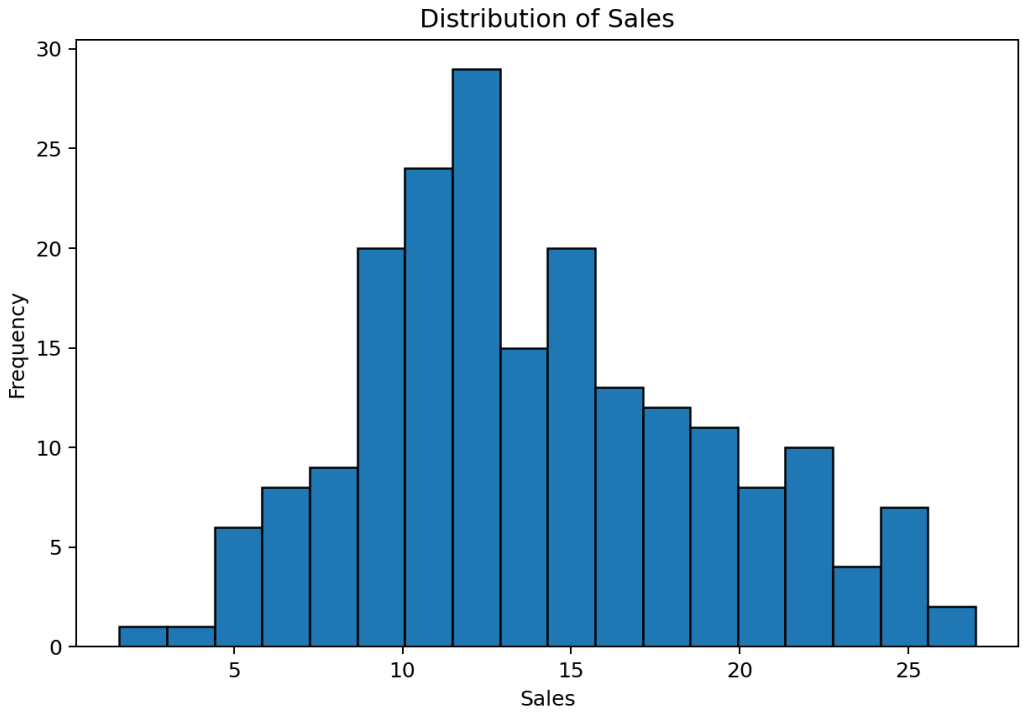
The original dataset contains 200 records and five columns. The index-like Unnamed: 0 field is excluded from training. The supplied file contains no missing values and no duplicate records. TV, Radio, and Newspaper are input features; Sales is the target variable.

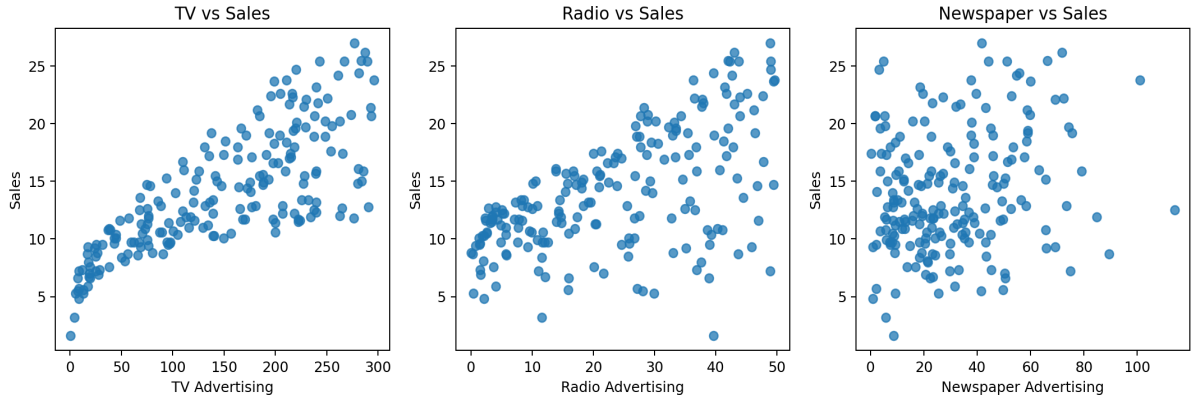
3. Data Preparation

The index-like field was removed. Missing-value and duplicate checks were performed. The data was divided into 80% training and 20% testing subsets using random_state=42.

4. Exploratory Data Analysis

The charts examine the distribution of sales and the relationships between each advertising channel and sales.





5. Model Development

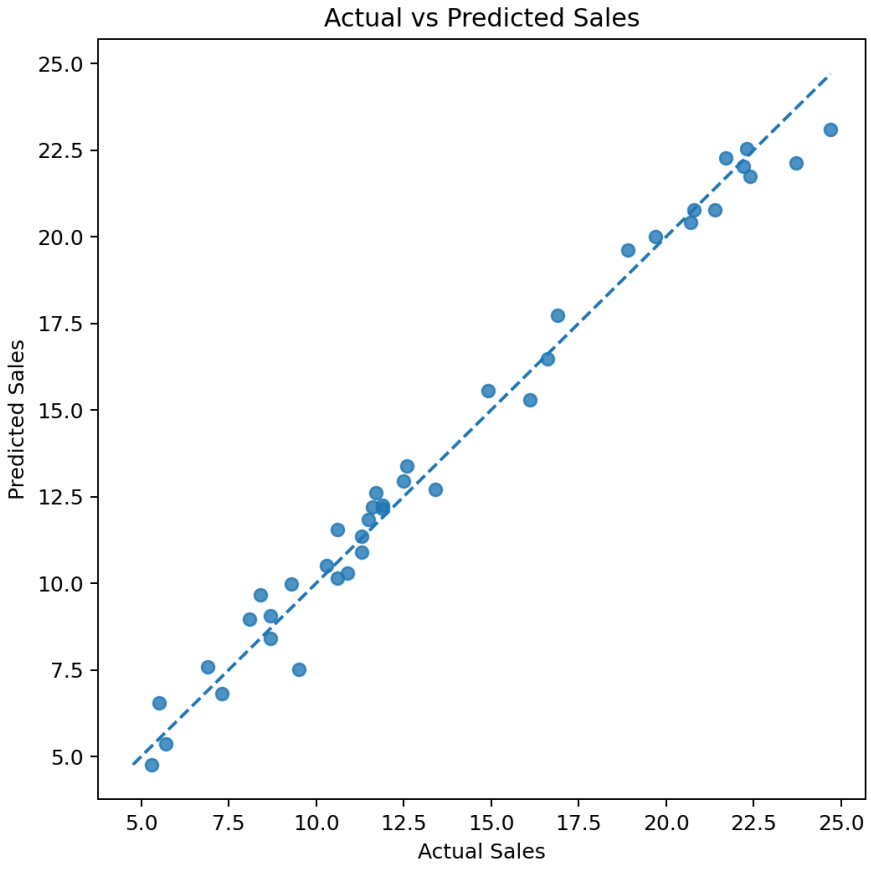
Linear Regression and Random Forest Regressor (400 trees) were trained using TV, Radio, and Newspaper advertising expenditure to predict Sales.

6. Model Evaluation

Models were compared using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2. Lower MAE and RMSE indicate smaller prediction errors; higher R^2 indicates that the model explains more variation in the test target.

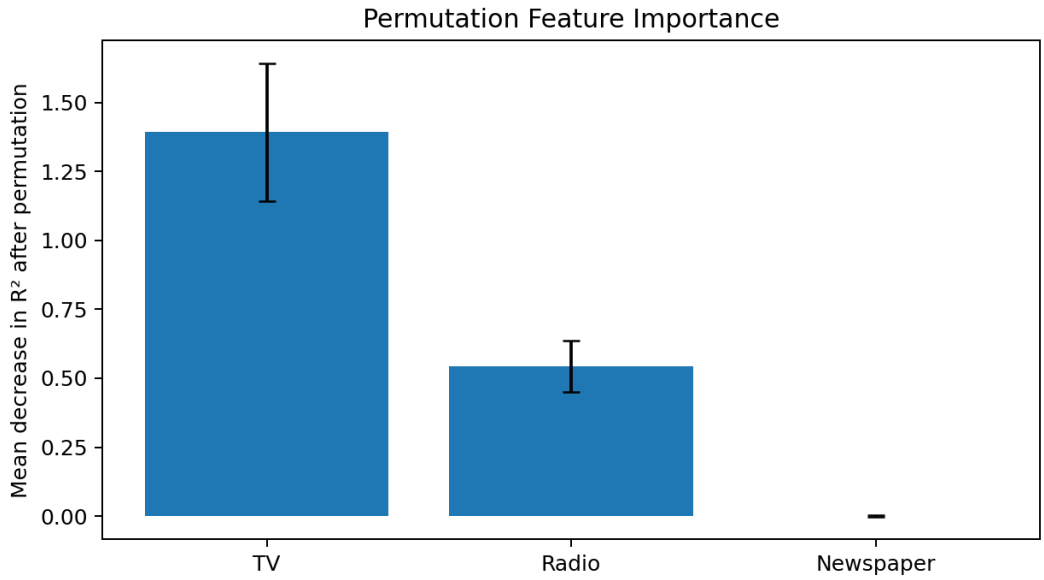
Model	MAE	RMSE	R^2
Random Forest Regressor	0.6210	0.7487	0.9822
Linear Regression	1.4608	1.7816	0.8994

The Random Forest Regressor model performed best on this fixed test split, with MAE 0.6210, RMSE 0.7487, and R^2 0.9822.



7. Feature Importance

Permutation importance measures the decrease in model score when a feature is shuffled. It is a model diagnostic and should not be interpreted as proof of causation.



8. Conclusion

The project demonstrates a complete regression workflow using Python and Scikit-learn. Random Forest Regressor performed best on the selected test split. Metrics, predictions, feature-importance output, notebook, scripts, and charts are included for reproducibility.

9. Limitations

The dataset is small and includes only advertising expenditure and sales. Other factors may affect sales, including pricing, seasonality, competition, product characteristics, and distribution. The model identifies patterns in the supplied dataset and does not establish that advertising expenditure causes sales changes. Validate using additional data before operational use.

10. Internship Information

Organization: CodeAlpha
Domain: Data Science
Task: Task 4 – Sales Prediction using Python